

**Network Pharmacology and Molecular Docking Analysis of Kochiae Fructus  
against Allergic Contact Dermatitis**



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## ABSTRACT

Allergic Contact Dermatitis (ACD) is a skin problem caused by inflammation, stress in skin cells, and an overactive immune response. Corticosteroids are commonly used for the treatment of ACD; however, their long-term use is associated with several adverse effects, highlighting the need for safer alternative therapies. Although *Kochiae Fructus* has long been used in traditional medicine to treat different skin diseases, scientists still do not fully understand how it works to help with ACD. Thus, the present work aimed to investigate therapeutic potential and the molecular mechanism of *Kochiae Fructus* anti-ACD activity using an integrated network pharmacology and molecular docking approach. The phytochemicals of *Kochiae Fructus* were retrieved from the TCMSP database, and the ACD-related genes were retrieved from the GeneCards database. Venny was used to identify the common targets and Protein-Protein Interaction (PPI) network was constructed using STRING and Cytoscape. Finally, a total of 211 intersecting targets were identified between the phytochemicals of *Kochia Fructus* and the genes related to ACD. A major hub gene related to inflammatory and immune pathways were identified as MAPK1, MAPK3, AKT1, JUN, STAT3, TP53, BCL2L1, CASP3, BCL2 and PIK3CB. GO and KEGG enrichment analysis were used to find out the significant biological processes and signaling pathways. To find the major pathway related to oxidative stress and inflammatory response, we performed KEGG pathway enrichment analysis, which revealed the AGE-RAGE signaling pathway. Molecular docking analysis using PyRx was performed to evaluate the binding interactions between the selected phytochemicals and the key target proteins involved in inflammation and apoptosis. MAPK3, BCL2, STAT3, AKT1 and CASP3 were determined, and the results revealed that Oleanolic acid had excellent binding affinities to all of the above molecules, which implies that it has a multi-target anti-inflammatory and cytoprotective effect. Overall, the findings provide computational evidence that *Kochiae Fructus* may act as a potential multi-target therapeutic agent against Allergic Contact Dermatitis through modulation of oxidative stress, MAPK-mediated inflammatory signaling pathways, and immune responses. These findings may serve as a basis for future experimental and clinical investigations.

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# **Chapter 1 - INTRODUCTION**

## **1.1 Overview**

Allergic Contact dermatitis, also called contact eczema, is a common skin problem where the skin becomes red and inflamed. It makes up about 20% of dermatitis cases in children [1]. The condition develops when individuals who have become sensitized are exposed to environmental allergens. It happens when a person becomes sensitive to certain things in the environment and then touches them again later. In areas of loose connective tissue, such as the eyelids, lips, and genital regions, diffuse swelling with poorly defined borders and altered skin texture can be seen. Although the reaction typically remains confined to the site of exposure, allergens may spread through scratching or systemic absorption, leading to lesions at distant sites. Acute symptoms often resolve after removal of the allergen; however, repeated or prolonged exposure can result in chronic skin changes, including thickening and lichenification [2]. In some cases, ACD may resemble other dermatological conditions, such as atopic dermatitis, or present in non-eczematous forms including erythema multiforme-like or pigmentary changes [3].

## **1.2 Pathophysiology of ACD**

At the molecular level, Allergic Contact Dermatitis is a delayed allergic reaction caused by T-cells. It happens in two stages: sensitization and elicitation. In the sensitization phase, the body first comes into contact with an allergen, which activates immune cells and T-cells. Upon repeated exposure, the elicitation phase is initiated, leading to an inflammatory reaction.

## **1.3 Role of T Cells and Cytokines in ACD**

Both CD4<sup>+</sup> and CD8<sup>+</sup> T cells play significant roles in the development of Allergic Contact Dermatitis [4], [5]. Among the different immune cells involved, CD8<sup>+</sup> T are mainly associated with inflammatory reactions, whereas CD4<sup>+</sup> may function in either pro-inflammatory or immunoregulatory manner depending on the surrounding cytokine milieu. In addition, cytokines including interferon-gamma (IFN- $\gamma$ ) and interleukin-17A (IL-17A) play important role in immune regulation and are closely related to Th1 and Th17 immune pathways [6].

## **1.4 Allergen-Specific Immune Responses**

Research conducted in humans has shown that immune responses in ACD can vary according to type of the allergen involved. Metal allergens, particularly nickel, are known to mainly stimulate Th1, Th17, and Th22 immune pathways [7]. In contrast, allergens such as fragrances and rubber compounds are generally associated with a Th2 domain response [8]. These differences demonstrate the complex nature of immune regulation in ACD and emphasize the need to identify the molecular targets associated with disease development and progression.

## **1.5 Therapeutic Potential of Kochiae Fructus**

Kochiae Fructus is derived from the dried fruit of “*Kochia scoparia*” and has long been used in traditional Chinese medicine for the treatment of different skin diseases such as eczema, dermatitis, pruritus, and psoriasis [9]. Phytochemical investigations have revealed the presence of several bioactive compounds in Kochiae Fructus. Many of these compounds are known to exhibit anti-inflammatory, anti-allergic, and protective effects on the skin. Although it has been used therapeutically for many years, the exact compounds responsible for its beneficial effects against ACD, along with their molecular mechanisms, have not yet been fully clarified.

## **1.6 Research Gap and Rationale**

The traditional clinical applications of Kochiae Fructus include treating a wide range of inflammatory skin conditions, but its mechanism of action in the treatment of Allergic Contact Dermatitis (ACD) is still not well understood. Its anti-inflammatory and immunomodulatory properties have not been systematically studied in terms of the key bioactive phytochemicals, molecular targets, hub genes and signaling pathways involved. Furthermore, the multi-component and multi-target mechanisms of Kochiae Fructus in ACD have not been fully explored using network pharmacology and molecular docking approaches. As long-term corticosteroid treatment may have some negative consequences, it is necessary to investigate safer alternatives from plants as a therapeutic option. Hence, this study was carried out to elucidate the molecular mechanisms underlying the anti-ACD activity of Kochiae Fructus.

## **1.7 Aim of the Study**

Accordingly, this study was designed to identify the major bioactive phytochemicals of Kochiae Fructus and investigate their potential molecular targets, hub genes, and signaling pathways involved in the treatment of Allergic Contact Dermatitis using integrated network pharmacology and molecular docking approaches.



## **Chapter 2 – MATERIALS & METHODS**

### **2.1 Tools and Databases**

Different Online tools and databases were used to study the phytochemicals of *Kochiae Fructus*, their targets, and genes related to Allergic Contact Dermatitis. The software and databases used are listed in Table 1, and the complete workflow is shown in Figure 1.

### **2.2 Phytochemical Collection**

The phytochemicals of *Kochiae Fructus* were collected from the Traditional Chinese Medicine Systems Pharmacology Database (TCMSP, Version 3.0) [10]. A total of 19 phytochemicals were identified.

### **2.3 Toxicity Analysis**

The Toxicity of Phytochemicals was obtained from ProTox (Version 3.0) [11].

### **2.4 Target Prediction for phytochemicals**

The possible biological targets of the phytochemicals were identified using the SwissTarget Prediction database [12]. Only targets with a probability score of  $> 0$  were selected for further study.

### **2.5 ACD Gene Search**

Genes related to ACD were collected from GeneCards using the keyword “allergic contact dermatitis” [13].

### **2.6 Common Target Analysis**

The common genes between *Kochiae Fructus* and ACD-related genes were identified using Venny [14].

### **2.7 PPI Network Study**

The protein-protein interaction network of the common target genes was generated using the STRING database with a highest confidence score (0.900) and limited to *Homo sapiens* and the disconnected nodes were removed from the network [15]. Subsequently, the PPI network file in the form of tab-separated values (tsv.) format was imported into Cytoscape software (version 3.10.4) [16]. The ‘Analyze Network’ tool was used to study the network and identify the main anti-ACD target genes based on degree centrality (DC) values. These genes were double-checked using CytoHubba plug-in.

## 2.8 MCODE Cluster Analysis

The MCODE plug-in in Cytoscape was used to find important clusters in the protein–protein interaction (PPI) network of 210 possible anti-ACD targets. The analysis was carried out using the following setting: degree cutoff = 2, node score cutoff = 0.2, k-core value = 2, and maximum depth = 100.

*Table 1 Software and webtools used for network pharmacology*

| Name                      | Link                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| TCMSP                     | <a href="https://tcmssp-e.com/tcmssp.php">https://tcmssp-e.com/tcmssp.php</a>                                           |
| Gene Cards                | <a href="https://www.genecards.org/">https://www.genecards.org/</a>                                                     |
| ProTox                    | <a href="https://tox.charite.de/prottox3/">https://tox.charite.de/prottox3/</a>                                         |
| Venny                     | <a href="https://bioinfogp.cnb.csic.es/tools/venny/">https://bioinfogp.cnb.csic.es/tools/venny/</a>                     |
| Swiss Target Prediction   | <a href="https://swisstargetprediction.ch/">https://swisstargetprediction.ch/</a>                                       |
| STRING                    | <a href="https://string-db.org/">https://string-db.org/</a>                                                             |
| Cytoscape                 | <a href="https://cytoscape.org/download.html">https://cytoscape.org/download.html</a>                                   |
| MCODE plug in (Cytoscape) | <a href="ftp://ftp.mshri.on.ca/pub/BIND/Tools/MCODE">ftp://ftp.mshri.on.ca/pub/BIND/Tools/MCODE</a>                     |
| Cytohubba Plug-in         | <a href="https://apps.cytoscape.org/apps/cytohubba">https://apps.cytoscape.org/apps/cytohubba</a>                       |
| Chimera                   | <a href="https://www.cgl.ucsf.edu/chimera/download.html">https://www.cgl.ucsf.edu/chimera/download.html</a>             |
| Open BaBel                | <a href="https://openbabel.org/index.html">https://openbabel.org/index.html</a>                                         |
| PyRx                      | <a href="https://pyrx.sourceforge.io/">https://pyrx.sourceforge.io/</a>                                                 |
| DAVID                     | <a href="https://david.ncifcrf.gov/">https://david.ncifcrf.gov/</a>                                                     |
| Discovery Studio          | <a href="https://www.3ds.com/products/biovia/discovery-studio">https://www.3ds.com/products/biovia/discovery-studio</a> |
| RCSB PDB                  | <a href="https://www.rcsb.org/">https://www.rcsb.org/</a>                                                               |
| PubChem                   | <a href="https://pubchem.ncbi.nlm.nih.gov/">https://pubchem.ncbi.nlm.nih.gov/</a>                                       |
| SwissAdme                 | <a href="https://www.swissadme.ch/">https://www.swissadme.ch/</a>                                                       |
| Zinc                      | <a href="https://zinc.docking.org/">https://zinc.docking.org/</a>                                                       |
| Bioinformatics Platform   | <a href="http://www.bioinformatics.com.cn/">http://www.bioinformatics.com.cn/</a>                                       |
| Shinny Go                 | <a href="https://bioinformatics.sdstate.edu/go/">https://bioinformatics.sdstate.edu/go/</a>                             |
| AlphaFold                 | <a href="https://alphafold.ebi.ac.uk/">https://alphafold.ebi.ac.uk/</a>                                                 |

## 2.9 Genes–Compounds Network Analysis

To explore the complex relationships and potential therapeutic mechanisms between ACD-associated genes and the phytochemicals of *Kochiae Fructus*, an integrated network analysis was performed using Cytoscape software (version 3.10.4).

## 2.10 Enrichment analysis

Gene Ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis were performed on the 211 common gene targets using DAVID database [17],

[18]. The GO analysis included three categories: cellular components (CC), biological processes (BP) and molecular functions (MF). The most significantly enriched GO terms and KEGG pathways were visualized through bubble charts using an online bioinformatics platform.

## **2.11 Docking Analysis**

The protein and ligand structures were prepared using UCSF Chimera. Molecular Docking studies were carried out using PyRx to evaluate the interaction between the selected phytochemicals and target proteins [19].

### **2.11.1 Protein Preparation**

The protein structure of JUN(1A02), TP53(3LGL), STAT3(6NJS), CASP3(1PAU), MAPK3(PDB ID:9UW3), MAPK1(PDB ID:2OJG), Akt1(PDB ID:1H10), BCL2(PDB ID: 6O0K), BCL2L1(PDB ID: 3SPF), were retrieved from Protein Data Bank (PDB) database, whereas the PIK3CB structure (UniProt ID:P42338) was obtained from the AlphaFold Protein Structure Database due to the lack of a suitable experimentally determined structure for docking studies. The selected protein structures were imported into UCSF Chimera for protein preparation prior to molecular docking analysis. Initially, water molecules and co-crystallized ligands were removed from the protein structures to obtain a clean receptor model. The active binding pocket and interacting residues were identified using the zone selection method within 5.0 Å of the selected ligand atoms. Subsequently, Dock Prep was performed, where solvent molecules and non-complexed ions were deleted, while alternate locations were retained only for the highest occupancy conformations. Incomplete side chains were repaired using the Dunbrack 2010 rotamer library. Hydrogen atoms were then added to stabilize the protein structure, followed by the assignment of Gasteiger charges for proper charge distribution. The prepared proteins were finally saved and utilized for molecular docking studies in PyRx to evaluate the binding interactions between selected phytochemicals and target proteins.

### **2.11.2 Ligand preparation**

The ligands MOL002211 (11,14-eicosadienoic acid), MOL000675 (Oleic acid), MOL000263 (Oleanolic acid), MOL002038 (9E,12Z-octadecadienoic acid), and MOL001739 (Zoomaric acid) were retrieved from the TCMSP database. The ligand structure was prepared and optimized using UCSF Chimera before performing molecular docking analysis. The ligand file was imported into the software first and the Dock Prep tool was used to clean and prepare the structure. In this process, solvent molecules were removed to remove excess water molecules that could affect the accuracy of the docking. Alternate locations were modified by keeping only the highest occupancy atoms. Incomplete side chains were fixed using the Dunbrack 2010 rotamer library. Special residue corrections were also automatically applied to ensure structural consistency, such as selenomethionine (MSE) to methionine (MET), bromo-UMP (5BU) to UMP (U), methylselenyl-dUMP (UMS) to UMP (U) and methylselenyl-dCMP (CSL) to CMP (C).

Subsequently, hydrogen atoms and atomic charges were added to stabilize the molecular structure and prepare it for computation analysis. Charge assignment showed the net charge of ligand was + 0 indicating the overall neutral and stable molecular state for docking studies.

Following structure preparation, energy minimization was performed using the Minimize Structure tool in Chimera to optimize the molecular geometry, minimizing steric clashes and unfavorable bond interactions. The minimization was carried out with 100 steepest descent steps with a step size of 0.02 Å followed by 10 conjugate gradient steps with a step size of 0.02 Å and an update interval of 10. In the minimization, no atoms were fixed and the ligand was optimized to full structural precision. This energy minimization step improved the stability and accuracy of the ligand conformation which can be used for reliable molecular docking and further bioinformatics analysis.

### **2.11.3 Grid Box setup**

The prepared protein structure was imported into PyRx platform and converted into macromolecule for docking analysis. The selected phytochemical ligands were individually prepared by using the Open Babel module in PyRx.

The search space for ligand binding for molecular docking was defined by creating a grid box around the active binding site of the target protein. The dimensions and the center coordinates of the grid box (X, Y and Z axes) were modified to sufficiently cover the residues of the active site and allow flexible ligand interaction in the binding pocket.

The grid box parameters were chosen based on the active site region of the protein to predict accurately the protein–ligand interactions. Following the grid box optimization, docking simulations were performed with the AutoDock Vina algorithm in PyRx.

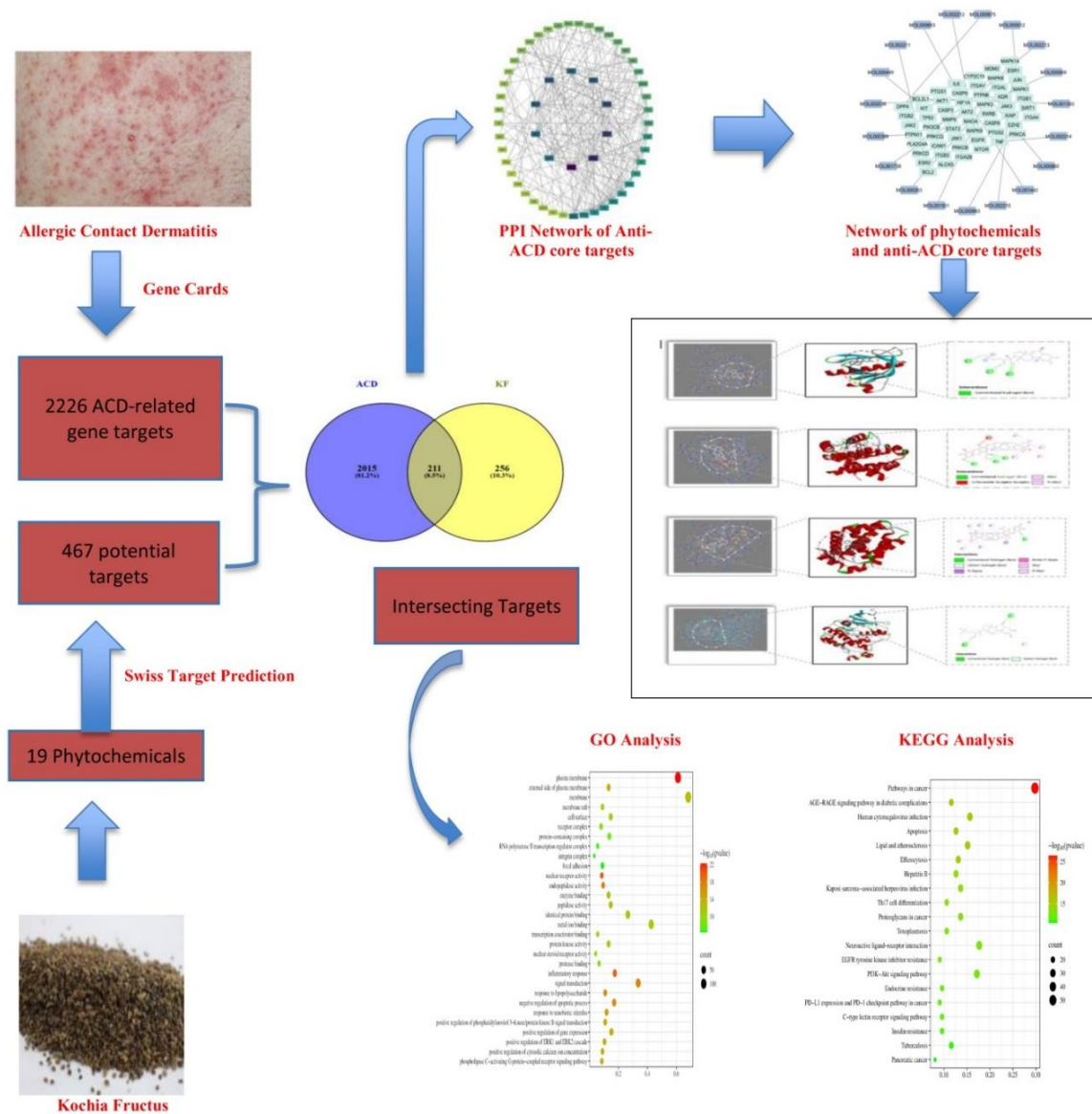


Figure 1 Workflow of methodology

## Chapter 3 – RESULTS

In this chapter the complete workflow proposed in Chapter 2 was systematically performed and evaluated step by step using a network pharmacology approach. All major computational analysis, including target identification, network construction, and molecular docking, were carried out to validate the proposed methodology and assess the overall performance of the study.

### 3.1 Target Screening of *Kochiae Fructus*

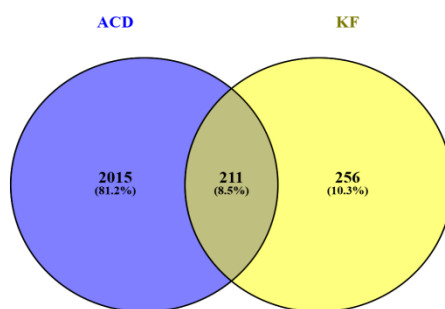
A total of 1861 possible targets of 19 phytochemicals were identified with a probability score of greater than 0. After removing repeated targets, 467 unique targets remained for further analysis.

### 3.2 ACD-Related Gene Search

To identify genes associated with Allergic Contact Dermatitis (ACD), the keyword “Allergic contact dermatitis” was searched in the GeneCards database. A total of 2226 ACD-related genes were retrieved. These genes were considered important for understanding the mechanisms of ACD and for investigating the therapeutic effects of *Kochiae Fructus* phytochemicals against the disease.

### 3.3. Common Target Analysis

The common targets between the phytochemical targets and the ACD-related genes were identified using Venny. A total of 211 intersecting gene targets were found and selected for further study, as shown in Figure 2. The common genes were considered as potential anti-ACD key targets.



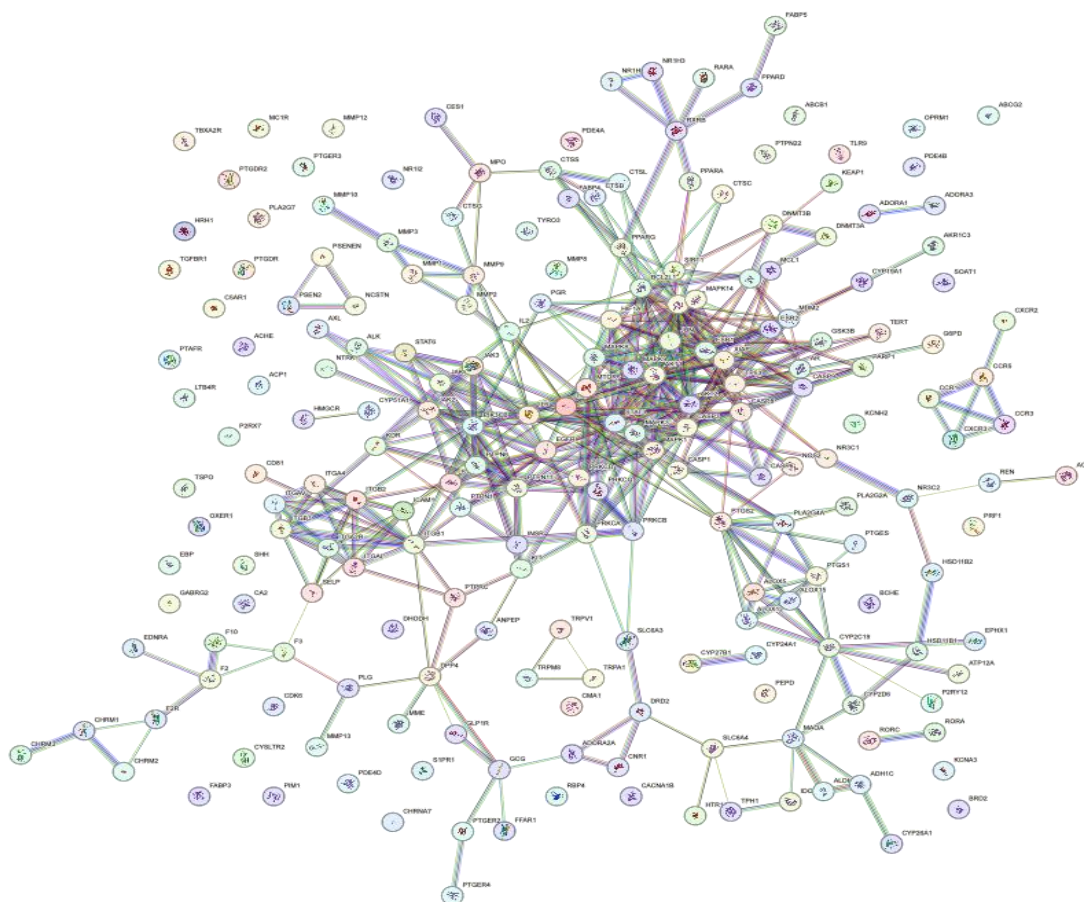
*Figure 2 Intersecting gene targets*

### 3.4. PPI Network Study of Common Targets

The 211 possible anti-ACD target genes were uploaded to the STRING database version 12.0 to construct the protein-protein interaction network, as shown in Figure 3. The results demonstrated that the network comprised nodes 210 and 452 edges. The average node degree, average local clustering coefficient, and average PPI enrichment p-values were 4.3, 0.469, and  $p < 1.0\text{e-}16$ , respectively.

Cytoscape analysis indicated that the PPI network constructed from STRING data contained 210 nodes and 452 edges. After importing and processing the network in cytoscape and performing network analysis, 49 disconnected or weakly connected nodes were removed, resulting in a refined network comprising 161 nodes and 452 edges with a characteristic path length of 4.162 between node pairs. Furthermore, the density, diameter, average number of neighbors, clustering coefficient, network heterogeneity, network centralization, and network radius are 0.043, 12, 6.127, 0.409, 0.906, 0.157, and 6, respectively.

A degree centrality (DC) threshold of  $> 6$  was applied to filter the network nodes. Based on this standard, 55 nodes were identified as the main potential targets of *Kochiae Fructus* for the treatment of Allergic Contact Dermatitis, as shown in Figure 4. Subsequently, the CytoHubba plugin in Cytoscape was applied to the PPI network to identify the top 10 hub genes. The selected 10 hub genes were then used to construct a hub gene network to visualize their key regulatory roles and interactions, as shown in Figure 5.



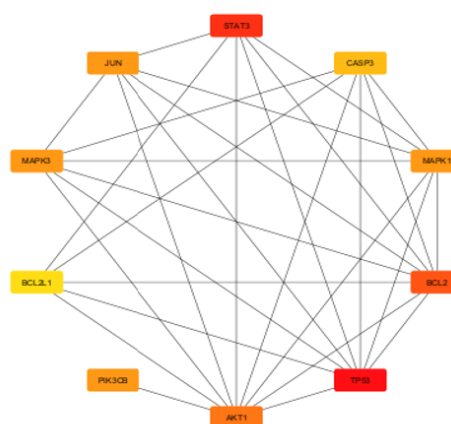
*Figure 3 String protein-protein interaction (PPI)*





*Figure 4 Network filtering (DC > 6)*

| Rank | Node   |
|------|--------|
| 1    | TP53   |
| 2    | STAT3  |
| 3    | BCL2   |
| 4    | AKT1   |
| 5    | MAPK1  |
| 5    | MAPK3  |
| 5    | JUN    |
| 5    | PIK3CB |
| 9    | CASP3  |
| 10   | BCL2L1 |



*Figure 5 Top Ten Targets and Network*

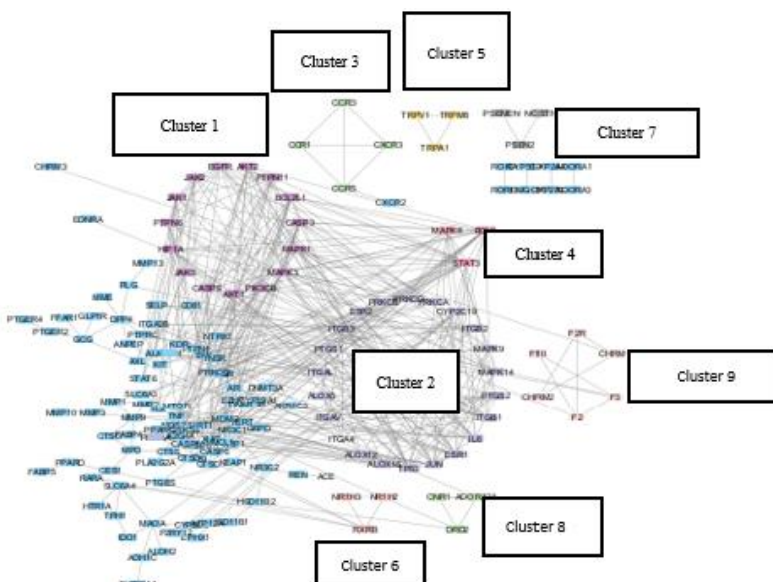
### 3.5. MCODE Cluster Study

The PPI network, which included 210 possible anti-Allergic contact dermatitis (anti-ACD) target proteins, was further analyzed using the MCODE plug-in in Cytoscape to identify important clusters. As shown in Figure 6, this analysis divided the network into nine different clusters. Cluster 1 contained 15 nodes and 47 edges and had a score of 6.714. Cluster 2 included 22 nodes and 52 edges with a score of 4.952. Clusters 3 contained 4 nodes and 6 edges with a score of 4.000. Cluster 4, 5, 6, 7 and 8 were smaller, and each had 3 nodes and 3 edges with a score of 3.000. Cluster 9 contained 6 nodes and 7 edges and had a score of 2.800. In Cluster 1, the proteins AKT1, MAPK3, MAPK1, PIK3CB, BCL2L1, PTPN11, AKT2, EGFR, JAK1, JAK2, PTPN6, JAK3, HIF1A, and CASP9 were strongly connected to several other targets, and each showed a degree value of 21, 19, 19, 19, 18, 17, 15, 14, 14, 13, 13, 12, 11, 11, 10. In Cluster 2, TP53, JUN, ESR1, IL6, ITGB1, PTGS2, MAPK14, MAPK9, CYP2C19, PRKCA, ITGB2, PRKCD, PRKCB, ITGB3, ESR2, ITGAL, PTGS1, ALOX5, ITGAV, ITGA4, and ALOX15 also showed strong interactions. Their degree values were 28, 19, 17, 15, 14, 13, 11, 11, 10, 10, 10, 9, 9, 8, 8, 7, 7, 6, 6, 6, 5, and 5 respectively. In Cluster 3, CCR5, CCR3, CCR1, and CXCR3, each having a degree value of 4, 3, 3, and 3. In Cluster 4, STAT3, BCL2, and MAPK8 and in Cluster 5, TRPM8, TRPA1,



and TRPV1 showed the same level of connectivity. All these nodes had a degree value of 2. In Cluster 6, RXRB, NR1H2, and NR1H3 with Degree values 6, 2, and 2. In Cluster 7, PSENEN, NCSTN, and PSEN2 with Degree value 2 and in Cluster 8, DRD2, ADORA2A, and CNR1 with Degree value, 4,3, and 2. In Cluster 9, F3, F2, CHRM1, F2R, CHRM2, and F10.

Importantly, Clusters 1 and 2 supported the identification of the ten major anti-ACD core targets: TP53, STAT3, BCL2, AKT1, MAPK1, MAPK3, JUN, PIK3CB, CASP3, and BCL2L1 as presented in Figure 6. The agreement between the cluster analysis and the PPI network results confirms that these key targets are reliable and may play an important role in the treatment of Allergic Contact Dermatitis.



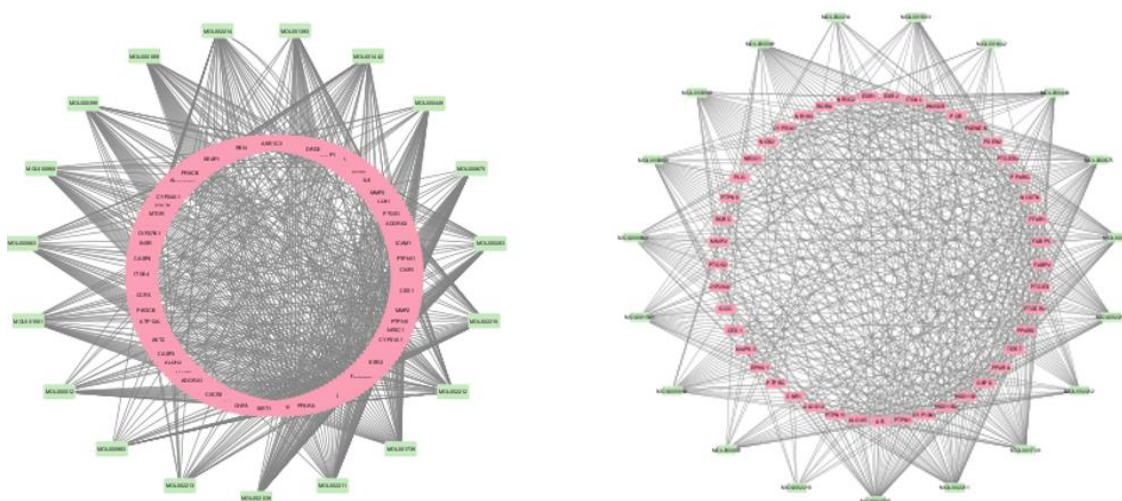
*Figure 6 Sub-network clusters generated by MCODE analysis*

### 3.6. Phytochemical-Target Network

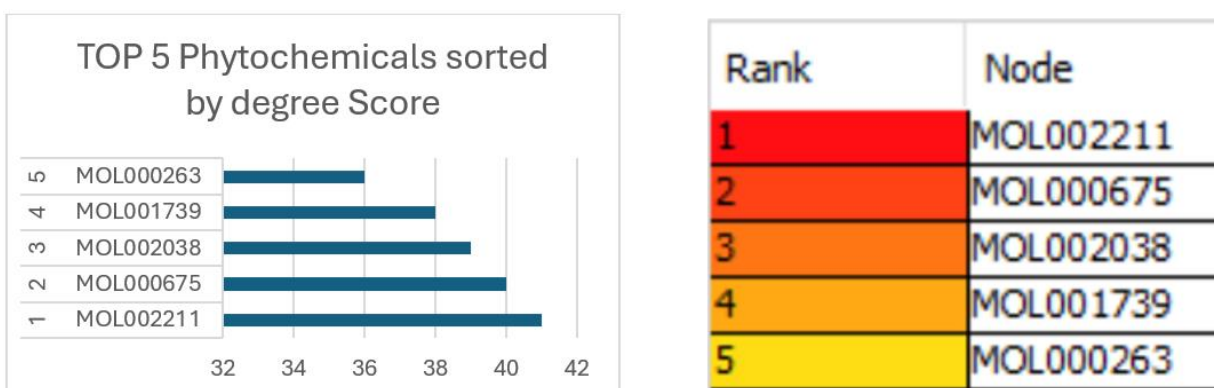
A network map was constructed using cytoscape to illustrate the interactions between the target genes and the MOLID's of the phytochemicals of *Kochiae Fructus*, as shown in Figure 7. This network contained 178 nodes and 788 edges. In this map, each edge represents a biochemical interaction between *Kochiae Fructus* phytochemicals and the anti-ACD targets.

The genes and phytochemicals interaction network was filtered using degree centrality to reduce the number of genes and phytochemicals in the network, thereby obtaining a refined network of 64 nodes with 529 edges as illustrated in Figure 7. The CytoHubba plugin in Cytoscape was used to find the most important hub genes and phytochemicals using network topology parameters. Based on this analysis, the most significant phytochemicals were identified, and a network was created to show the most influential interactions in the system as shown in Figure 8.

Network analysis demonstrated that all selected phytochemicals interacted with the core targets to varying extents, indicating a multi-component and multi-target therapeutic mechanism. Among the analyzed compounds, MOL002211 (11,14-eicosadienoic acid), MOL000675 (Oleic acid), MOL000263 (Oleanolic acid), MOL002038 (9E,12Z-octadecadienoic acid), and MOL001739 (Zoomaric acid) exhibited higher degree centrality (DC) values, suggesting stronger interactions with multiple core targets in the rank. Similarly, MAPK3, MAPK1, AKT1, BCL2L1, BCL2, JUN, PIK3CB, STAT3, TP53, and CASP3 emerge as the major hub targets due to their elevated connectivity and central positions within the network.



*Figure 7 Network between compounds and genes*



*Figure 8 Top 5 phytochemicals*

### 3.7. GO Analysis

The 211 possible anti-ACD key targets implicated in ACD were meticulously subjected to GO enrichment analysis. The outcomes enumerate the top 10 enrichment terms across the molecular function (MF), cellular component (CC), and biological process (BP) categories pertinent to these targets as shown in Figure 9. This analysis revealed that the anti-ACD key targets of *Kochia* Fructus phytochemicals within the BP category are intricately involved in an array of membrane-related and regulatory activities, including the plasma membrane, external side of plasma membrane, membrane raft, and cell surface indicating that these proteins are primarily involved in membrane-associated biological events. In addition, enrichment of receptor complex, protein-containing complex, and RNA polymerase II transcription regulatory complex suggests their role in signal reception and transcriptional regulation. The presence of focal adhesion-related enrichment further indicates involvement in cell communication and structural organization.

The cellular component (CC) enrichment analysis indicates that the identified proteins are predominantly localized in functional receptor and regulatory complexes. Significant enrichment was observed in nuclear receptor activity-related complexes, peptidase activity-associated structures, enzyme binding regions, and transcription coactivator binding components. The enrichment of protein kinase activity and nuclear steroid receptor activity suggests that many of the proteins are involved in intracellular signaling regulation and gene expression control.

Molecular function (MF) enrichment analysis highlights that the target proteins are mainly involved in signal transduction and regulatory molecular activities. The most enriched functions include inflammatory response, signal transduction, response to lipopolysaccharide, and negative regulation of apoptotic processes, indicating their role in immune and stress-related cellular responses. Additionally, enrichment in positive regulation of kinase activity, phosphatidylinositol signaling, ERK1/ERK2 cascade regulation, and calcium ion concentration regulation suggest active participation in intracellular signaling pathways. The presence of G protein-coupled receptor signaling pathway enrichment further supports their involvement in receptor-mediated cellular communication.

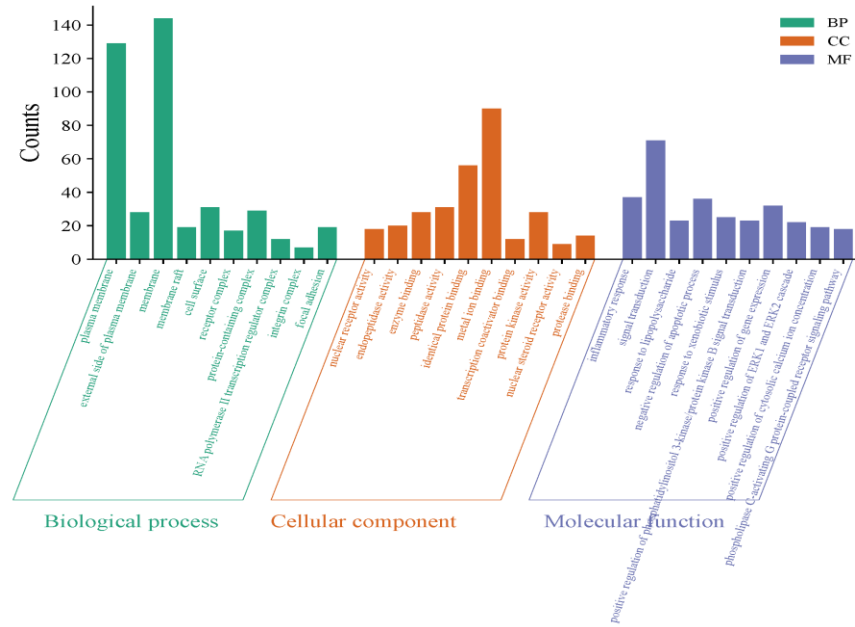


Figure 9 Top 10 GO enrichment analysis

### 3.8 KEGG pathway Study

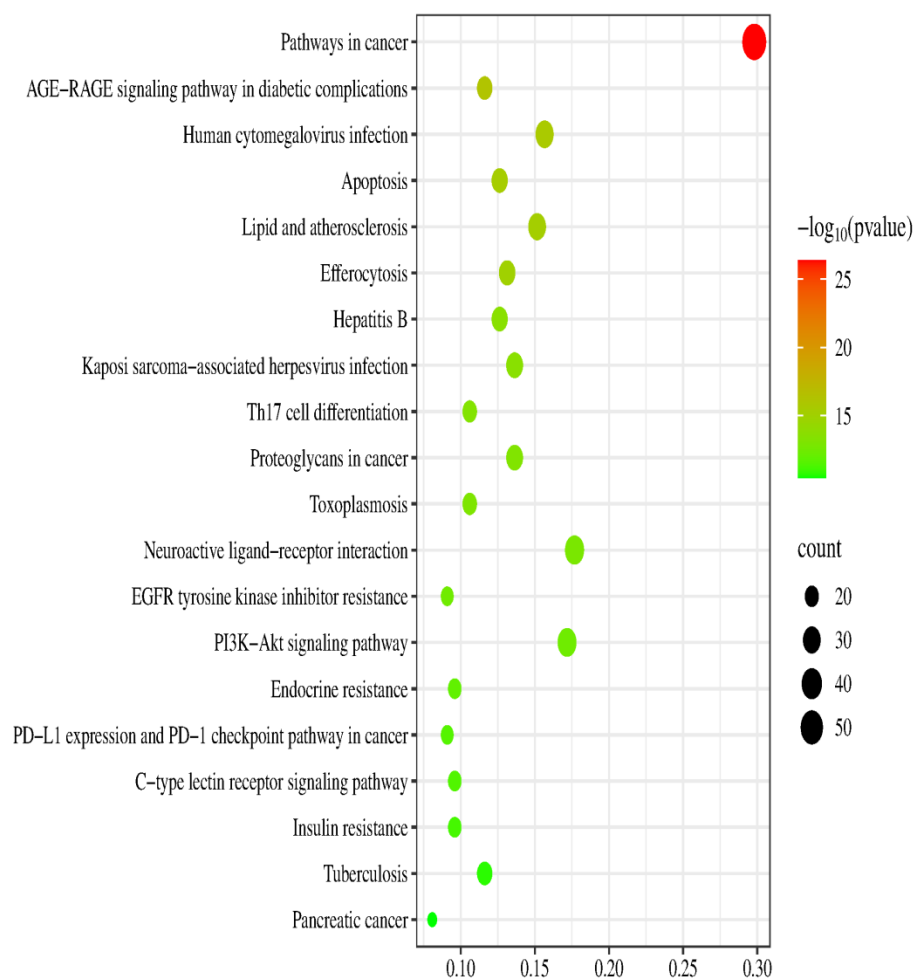
The KEGG pathway enrichment analysis identified the top 20 significantly enriched pathways associated with the selected target genes, as illustrated in the bubble plot. In Figure 10, the x-axis represents the gene ratio (the proportion of input genes enriched in each pathway), the bubble size indicates the number of genes involved (gene count), and the color gradient reflects statistical significance expressed as  $-\log_{10}(\text{p-value})$ , where darker red donates higher significance.

Among all pathways, Pathway in cancer shows highest gene ratio, the largest gene count and the strongest statistical significance, indicating that a substantial proportion of the identified targets involved in proliferation, survival, apoptosis, and inflammatory signaling networks. The PI3K-Akt signaling pathway was also prominently enriched, suggesting a central role in regulating cell survival, immune responses, and inflammatory process. Similarly, the enrichment of the AGE-RAGE signaling pathway highlights the involvement of oxidative stress and chronic inflammation. Immune-related pathways such as Th17 cell differentiation, PD-L1 expression and PD-1 checkpoint pathway in cancer, and C-type lectin receptor signaling pathway indicate that the identified genes participate in T-cell differentiation, cytokine production, and immune modulation. The presence of the Apoptosis pathway further suggests regulation of programmed cell death and tissue homeostasis.

Additionally, several infection-related pathways, including human cytomegalovirus infection, hepatitis B, Kaposi sarcoma-associated herpesvirus infection, tuberculosis, and toxoplasmosis, were significantly enriched, reflecting shared inflammatory and immune signaling components. Metabolic and resistance-related pathways such as lipid and atherosclerosis, insulin resistance,

endocrine resistance, and EGFR tyrosine kinase inhibitor resistance were also identified, suggesting links to metabolic inflammation and cellular stress responses.

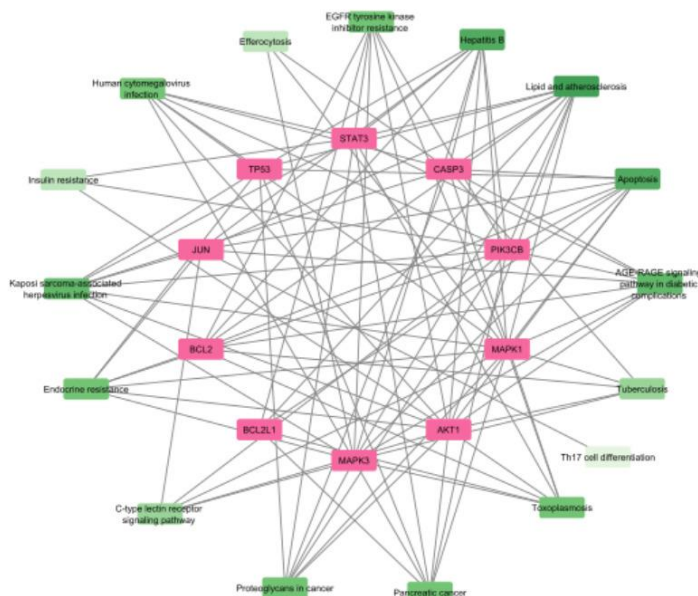
Overall, the KEGG enrichment results demonstrate that the target genes are mainly involved in cancer-related, immune-regulatory, inflammatory, apoptotic, and metabolic signaling pathways, indicating that their therapeutic effects may be mediated through modulation of key survival and immune signaling cascades, particularly PI3K–Akt and inflammation-associated pathways.



*Figure 10 Top 20 KEGG pathways*

### 3.9 Core pathway analysis

This network diagram represents the interaction between the top 20 enriched KEGG pathways and the top 10 hub genes identified through Cytoscape analysis. In Figure 11, the pink nodes represent hub genes such as TP53, STAT3, CASP3, MAPK1, MAPK3, BCL2, BCL2L1, JUN, PIK3CB, and AKT1, while the green nodes represent significant KEGG pathways. The connections (edges) between nodes indicate the involvement and interaction of these hub genes within multiple biological pathways. The analysis shows that the hub genes are highly interconnected and involved in important biological processes such as apoptosis, AGE-RAGE signaling pathway, cancer-related pathways, and immune responses.



*Figure 11 Network between hub genes and pathways*

### 3.10 Molecular Docking Analysis

Molecular docking was carried out between the top 10 anti-ACD core targets and top five phytochemicals of *Kochia fructus*. The docking outcomes were validated based on the binding affinity score, with more binding energy values having more binding interactions and more biological activity, as shown in Figure 12.

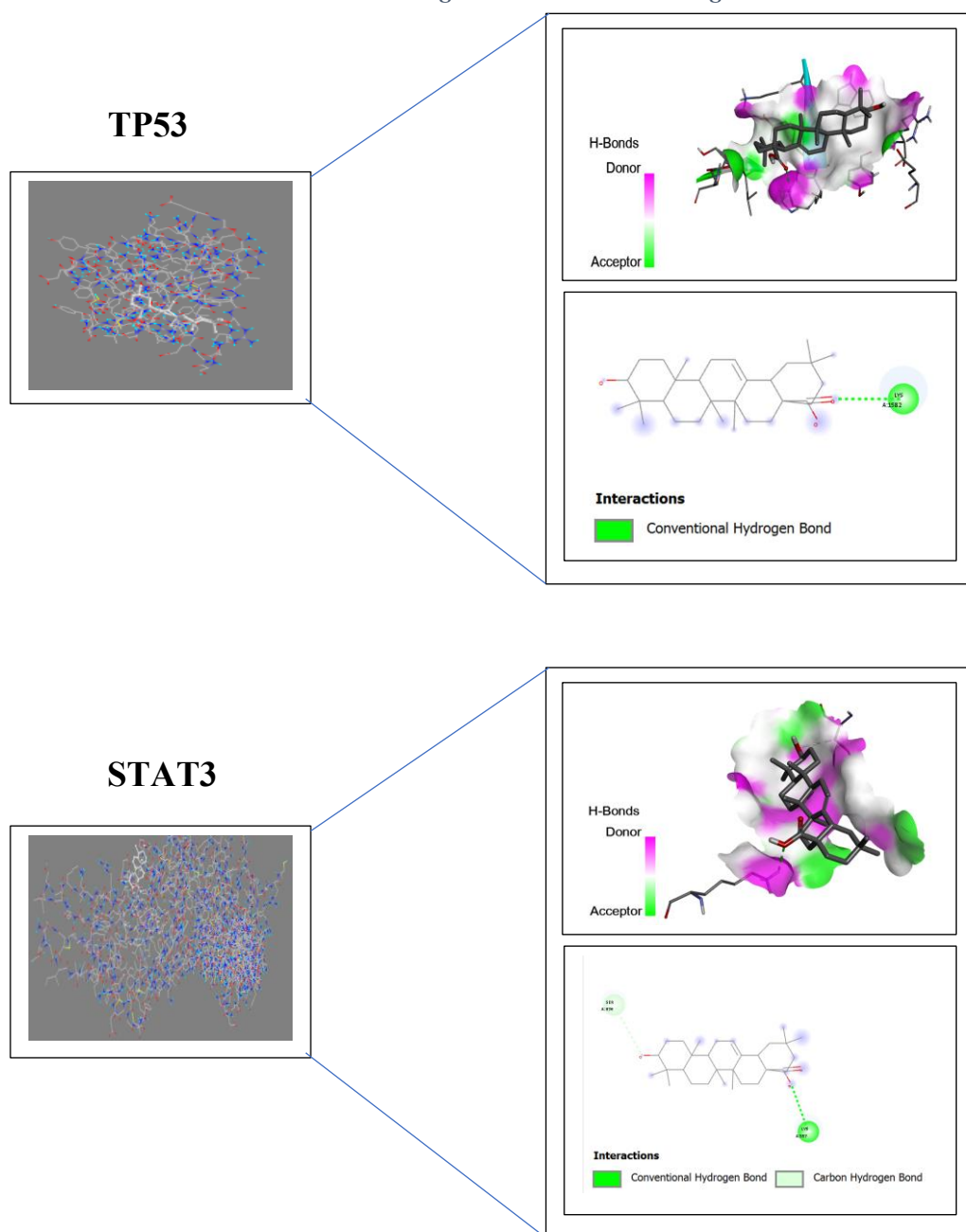
In addition, a number of complexes with high binding affinities (greater than -4 kcal/mol) were shown to be docked in Figure 13. Remarkably, among all the phytochemicals, oleanolic acid (MOL000263) had the highest binding affinity against the core targets: BCL2, (-8.3 kcal/mol), TP53, (-7.7 kcal/mol), MAPK3, (-7.2 kcal/mol), STAT3, (-7.0 kcal/mol), AKT1, (-6.8 kcal/mol), PIK3CB, (-6.8 kcal/mol), and CASP3, (-6.5 kcal/mol). Structurally, oleanolic acid formed key hydrogen bonds with CASP3 (H-Bond: THR177, ARG34, HIS237), TP53 (H-Bond: LYS1582), BCL2 (H-Bond: ASN143, ASN143), and AKT1 (H-Bond: GLU17, ARG23, GLN79, GLY16).



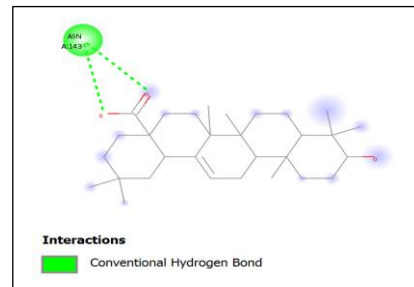
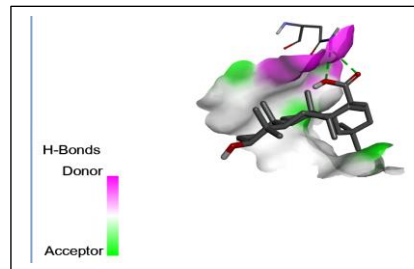
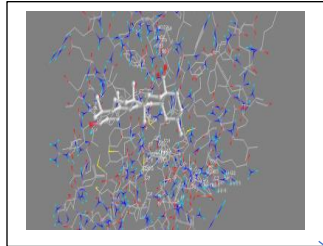
Besides, the oleanolic acid was interacted with MAPK1 (-6.4 kcal/mol) by hydrogen bond at LYS112, THR108, ASP109 and with MAPK3 (-7.2 kcal/mol) by hydrogen bond at LYS72. It also formed a hydrogen bond with PIK3CB (-6.8 kcal/mol) at ASP809. On the other hand, JUN (-4.7 kcal/mol) exhibited more van der Waals interactions with LYS268, ARG269, ASN272, and ARG273. The binding affinities of the other 4 phytochemicals of *Kochia fructus* were analyzed to compare with oleanolic acid (MOL000263), which exhibited the best overall binding affinities with the key hub targets. Thus, the complexes of this important phytochemical, docked together, can be further studied by MD simulation.

|                                        | TP53 | STAT3 | BCL2 | AKT1 | MAPK1 | MAPK3 | JUN  | PIK3CB | CASP3 | BCL2L1 |
|----------------------------------------|------|-------|------|------|-------|-------|------|--------|-------|--------|
| MOL002211(11,14-eicosadienoic acid)    | -5.7 | -4.9  | -6.4 | -4.1 | -6.3  | -5.3  | -3.3 | -5.5   | -4.5  | -5.9   |
| MOL000675(OLEIC ACID)                  | -5.7 | -4    | -6   | -3.9 | -6.4  | -5.3  | -3   | -5.6   | -4.2  | -5.9   |
| MOL000263(OLEANOLIC ACID)              | -7.7 | -7    | -8.3 | -6.8 | 1.4   | -7.2  | -4.7 | -6.8   | -6.5  | -7.6   |
| MOL002038(9E,12Z-octadecadienoic acid) | -5.9 | -4    | -6.3 | -4.2 | -6.3  | -5.6  | -2.8 | -5.8   | -4.7  | -5.9   |
| MOL001739(Zoomaric acid)               | -5.7 | -4.9  | -5.7 | -4.1 | -6    | -5.1  | -2.9 | -5.1   | -4.2  | -5.5   |

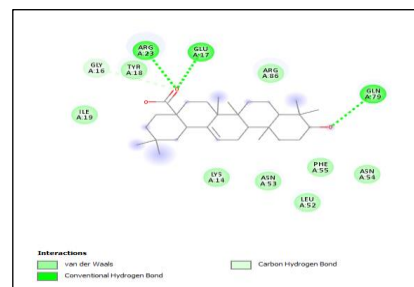
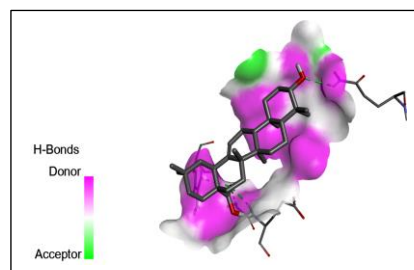
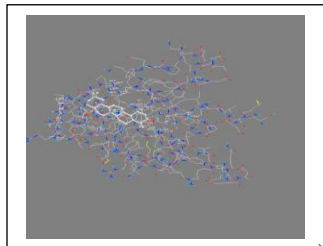
Figure 12 Molecular docking score



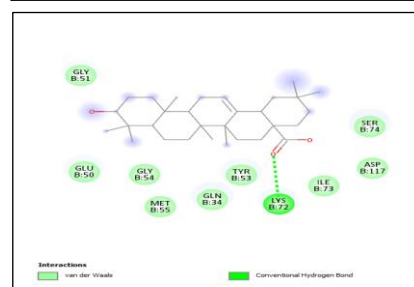
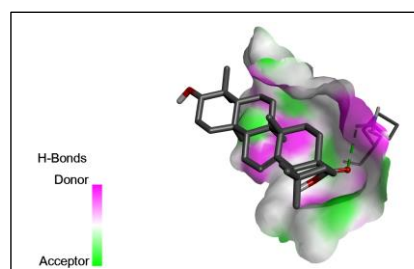
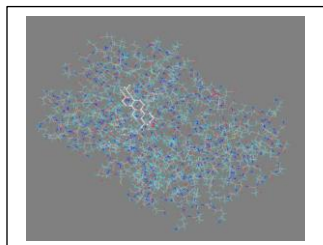
## BCL2



## AKT1

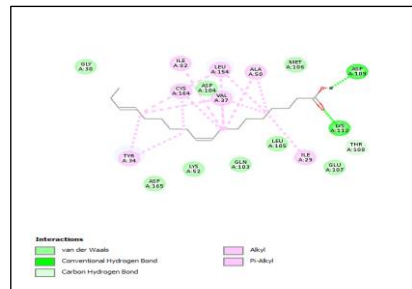
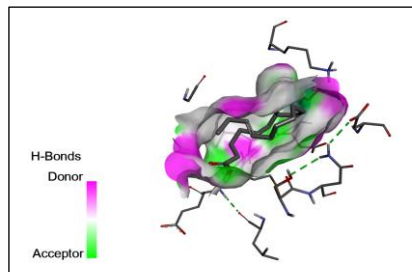
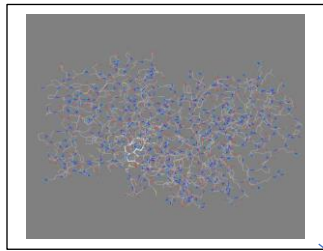


## MAPK3

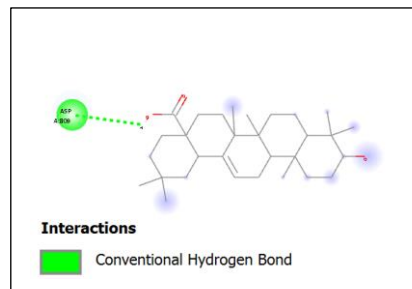
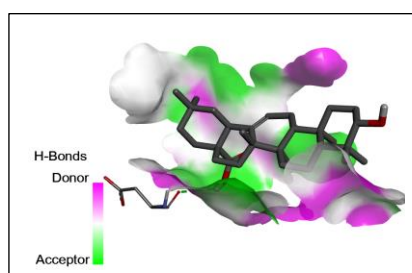
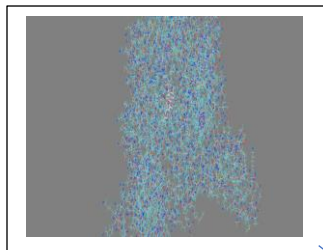




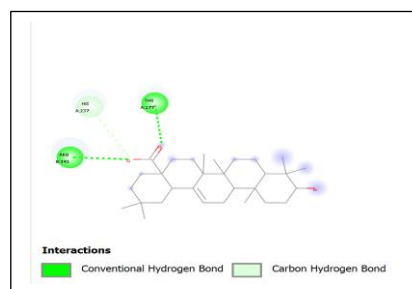
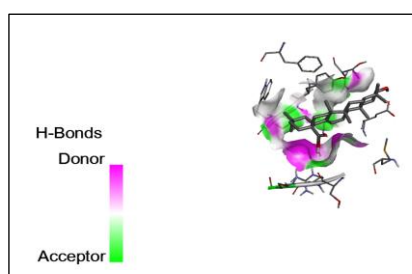
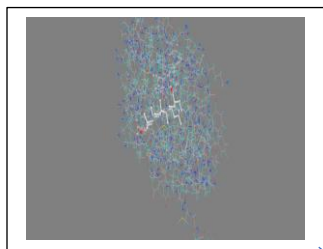
## MAPK1

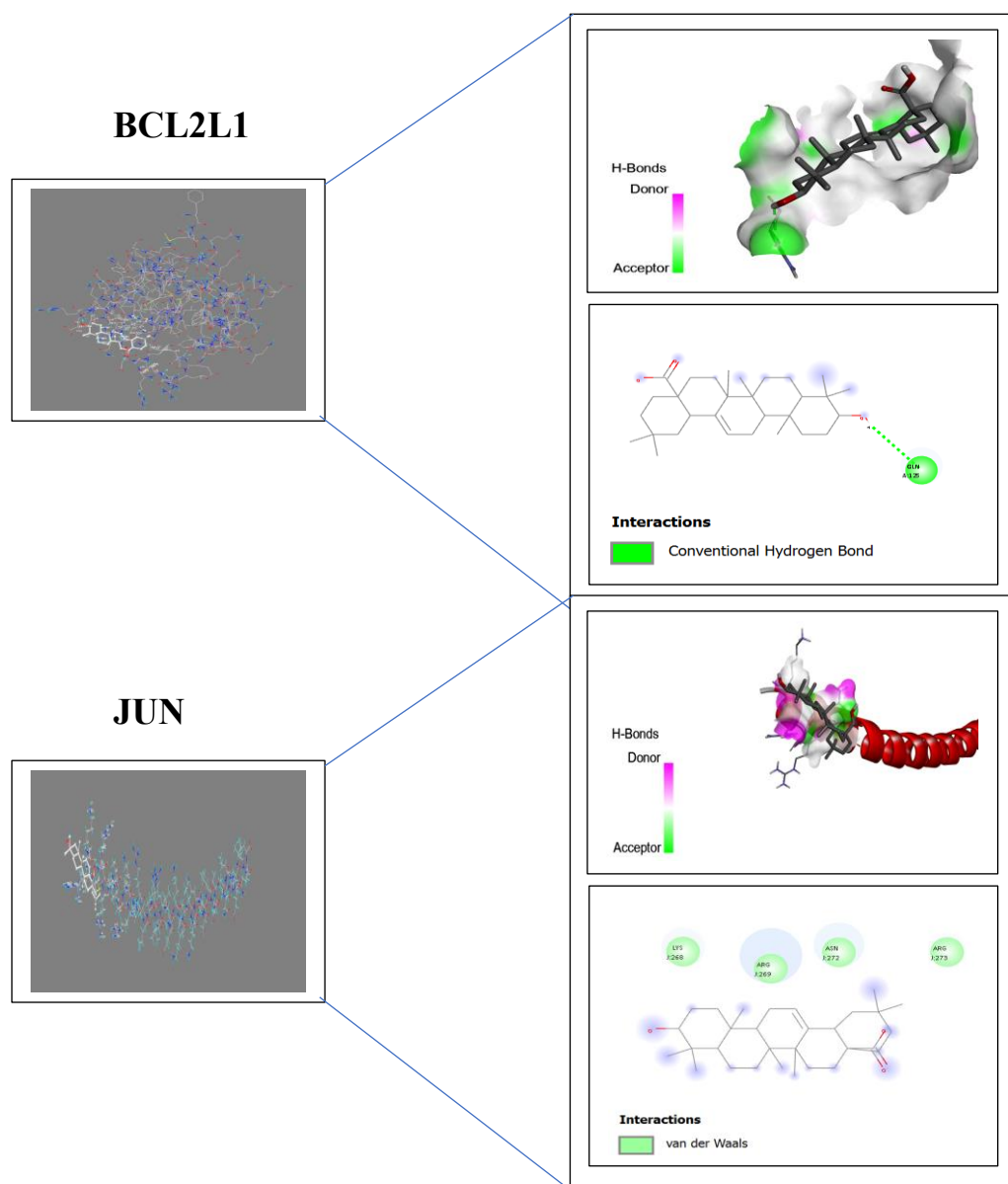


## PIK3CB



## CASP3





*Figure 13: Molecular docking results*

### 3.11 Pathway-Based Network Analysis

To investigate the molecular mechanism through which *Kochiae Fructus* may exert therapeutic effects against Allergic Contact Dermatitis, KEGG pathway enrichment analysis was performed. A total of 174 pathways were initially identified after submitting the potential anti-Allergic Contact Dermatitis targets of *Kochiae Fructus* to the DAVID bioinformatics platform. After applying a significance cutoff of  $p < 0.05$ , 42 significantly enriched pathways remained. Among these, the top 20 pathways provided by DAVID were selected for further evaluation. The AGE-RAGE signaling pathway was selected as the key pathway because it contained the highest number of hub genes among the enriched pathways. This pathway is strongly associated with inflammatory

responses, oxidative stress, and immune dysregulation, which are important pathological features of Allergic Contact Dermatitis.

As illustrated in Figure 14, the activation of the AGE-RAGE pathway is involved in a cascade of events that involve important signaling proteins such as AKT(AKT1), STAT3, and ERK1/2 (MAPK1/MAPK3). The proteins function as messengers to activate important controllers in the cell, called AP-1 and NF- $\kappa$ B. These active controllers trigger the cell to release a storm of inflammatory pro-inflammatory signals (IL-1, IL-6 and TNF- $\alpha$ ) that directly lead to severe inflammation and swelling of the skin. Meanwhile, this ongoing inflammation puts the skin cells under a tremendous amount of stress that forces them to enter into the process of cellular damage and programmed cell death (Apoptosis) that is regulated downstream by Bcl-2 and Caspase-3 (CASP3). Our molecular docking results clearly indicate that Oleanolic acid (MOL000263) has excellent binding affinities towards these same crossroad molecules: BCL2 (-8.3 kcal/mol), STAT3 (-7.0 kcal/mol), AKT1 (-6.8 kcal/mol) and CASP3 (-6.5 kcal/mol). Oleanolic acid can bind tightly to these core proteins and effectively block the chain of reactions that results in the production of inflammatory signals, while also shielding the skin cells from the apoptotic damage. As a result, the product effectively slows the progression of chronic inflammation and alleviates the damage of the tissue caused by ACD, especially by its main active ingredient Oleanolic acid. The results further confirmed the reliability of the network pharmacology method used in the study.

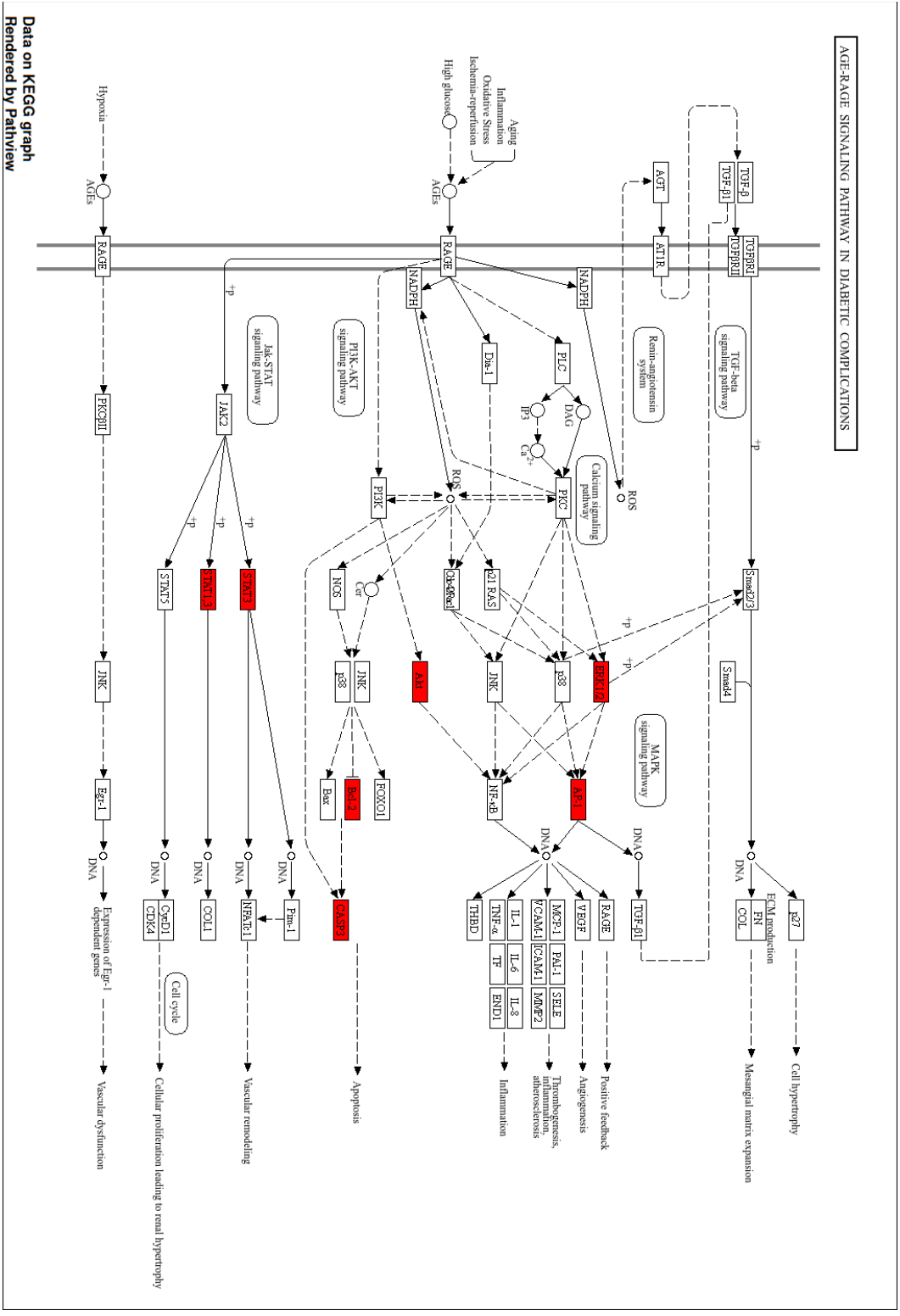


Figure 14 AGE-RAGE signaling pathway

## Chapter 4 – DISCUSSION & CONCLUSION

The Allergic Contact Dermatitis (ACD) is a T-cell mediated inflammatory skin disease, which corresponds to immune dysregulation, oxidative stress, cytokine release and activation of various inflammatory pathways. The current study used the network pharmacology and molecular docking method to explore the therapeutic potential and molecular mechanism of *Kochiae Fructus* in treating ACD. Based on integrated analysis, it has been found that *Kochiae Fructus* has a multi-component and multi-target therapeutic mechanism.

The number of intersecting targets between the phytochemicals of *Kochiae Fructus* and ACD-related genes was 211, which shows a high degree of association between the active constituents of the plant and the molecular mechanisms involved in the progression of ACD. The protein–protein interaction (PPI) network analysis also revealed a number of hub genes, including TP53, STAT3, AKT1, MAPK1, MAPK3, BCL2, BCL2L1, JUN, CASP3 and PIK3CB. The genes were well recognized for their role in inflammatory response, oxidative stress, apoptosis, immune signaling, and cell survival pathways, which indicated that they might have central roles in the anti-inflammatory activity of *Kochiae Fructus*.

These hub targets within the interaction network were further confirmed as important by the MCODE cluster analysis and CytoHubba validation. Specifically, the connectivity and degree centrality values of the MAPK1, MAPK3, AKT1, BCL2L1, TP53, STAT3, PIK3CB, JUN, CASP3 and BCL2 genes were higher, suggesting their important role in the therapeutic mechanism of *Kochiae Fructus*. These mutually dependent targets further validate the network pharmacology strategy employed in this research.

The targets identified by the GO enrichment analysis are significantly related to inflammatory responses, signal transduction, regulation of the apoptotic process, activity of a kinase, receptor signaling, and immune related biological functions. Furthermore, the cellular component analysis revealed enrichment of complexes associated with the cell's membrane and receptor-related structures, and the molecular function analysis revealed enrichment of protein kinase activity, enzyme binding, and transcriptional regulation. These results imply that *Kochiae Fructus* may have effect on the regulation of several biological pathways related to inflammatory skin diseases.

Pathway enrichment analysis showed that the pathways of inflammatory and immune responses were strongly enriched such as the PD-1/PD-L1 signaling pathway, the apoptosis pathway, the Th17 cell differentiation pathway, the AGE-RAGE signaling pathway, and the PI3K-Akt signaling pathway. These routes are closely linked to immune dysfunction, oxidative stress, production of inflammatory cytokines and chronic inflammatory responses related to the development of ACD.

In the enriched pathways, the AGE-RAGE pathway was identified as the highest ranked pathway due to the number of hub genes associated with oxidative stress and inflammation as well as cellular survival were present in the pathway. The AGE-RAGE pathway activates the downstream signaling cascades, leading to increase in production of intracellular reactive oxygen species

(ROS) that activate important hub molecules such as AKT(AKT1), STAT3 and ERK1/2 (MAPK1/MAPK3). These molecules subsequently interact with other pathways in the body to stimulate important transcription factors, including AP-1 and NF- $\kappa$ B, which increase the production of pro-inflammatory cytokines such as IL-1, IL-6 and TNF- $\alpha$ . These signaling molecules, working together, stimulate more inflammatory cytokines that contribute to continued inflammation and immune dysfunction in ACD. At the same time, cells in this highly inflammatory environment are undergoing a cell stress response, pushing them toward apoptotic damage and programmed cell death, which is regulated by Bcl-2 and Caspase-3 (CASP3). Pathway analysis suggested important interconnected factors contributing to the development and progression of Allergic Contact Dermatitis: Excessive ROS production, chronic inflammation and subsequent apoptotic tissue injury. Therefore, blocking this multi-axial inflammatory and apoptotic signaling may prove a promising therapeutic strategy to control tissue damage and inflammation in ACD. The selected targets in this cascade were further analyzed through molecular docking analysis with the dominating phytochemicals of the *Kochiae Fructus*. The results revealed very high binding affinities of the phytochemicals towards the target proteins. Interestingly, in contrast to previously analyzed compounds, Oleanolic acid (MOL000263) was most highly represented bioactive component with very strong and comparatively better binding interactions with the core targets BCL2 (-8.3 kcal/mol), STAT3 (-7.0 kcal/mol), AKT1 (-6.8 kcal/mol), and CASP3 (-6.5 kcal/mol). These interactions strongly indicate that Oleanolic acid might simultaneously inhibit the upstream inflammatory signaling pathways [AKT(AKT1), STAT3], and directly stabilize the downstream survival machinery (BCL2, CASP3), thus reducing inflammatory responses and cellular damages in ACD. Moreover, the docking complexes were found to be structurally safe and stable when the specific hydrogen bond interactions between the docked proteins and the ligand were seen, as hydrogen bonds with CASP3 (THR177, ARG34, HIS237), BCL2 (LYS551, SER636), and AKT1 (ASN143) were observed. The binding interactions of Oleanolic acid in these different inflammatory and apoptotic targets were very stable, suggesting that Oleanolic acid is a major bioactive compound responsible for the therapeutic effect of *Kochiae Fructus*.

There are currently several treatments available for Allergic Contact Dermatitis, such as corticosteroid-based creams and medicines. Although these medications are effective in reducing inflammation, prolonged use may lead to several side effects such as skin thinning, irritation, delayed wound healing, and immune suppression. Hence, plant-based therapeutic agents like *Kochiae Fructus* might be considered as relatively safer and multi-targeted agents to control ACD. In general, results of the present study are computational and supportive of the therapeutic application of the extracts of *Kochiae Fructus* on Allergic Contact Dermatitis. Overall, the results from network pharmacology, PPI network, GO and KEGG enrichment analysis, CytoHubba validation, MCODE clustering, and molecular docking studies indicate that *Kochiae Fructus* could potentially modulate oxidative stress, inflammatory signaling pathways, immune responses, and MAPK-associated pathways also other pathways to help stop the progression of ACD. The identified phytochemicals and target interactions need to be validated in a molecular dynamics

(MD) simulation, in vitro, and in vivo studies to prove the biological activity, safety, and therapeutic efficacy of the identified phytochemicals and target interactions.

## **Conclusion**

In the present study, the therapeutic potential of *Kochiae Fructus* against allergic contact dermatitis (ACD) was investigated through network pharmacology and molecular docking approaches, considering the limitations and adverse effects associated with conventional corticosteroid therapy. Key active phytochemicals, potential therapeutic targets and remarkably enriched signaling pathways were identified that may aid the utilization of *Kochiae Fructus* in ACD treatment. The results indicated that therapeutic activity of *Kochiae Fructus* was multi-component and multi-target including inflammatory response, oxidative stress, immune response and downstream apoptotic pathways. The AGE–RAGE signaling pathway was identified as a key pathway related to inflammation and oxidative stress in the enriched pathways in ACD. The molecular docking study also demonstrated that the active phytochemicals possess good binding affinities towards the core target proteins, including MAPK3, BCL2, STAT3, AKT1 and CASP3, demonstrating their anti-inflammatory and cytoprotective activities. Overall, the results from the present study can support the scientific basis for the traditional use of *Kochiae Fructus* in treating inflammatory skin disorders. Based on the bioactive compounds, key targets and signaling pathways identified, *Kochiae Fructus* could be a potential therapeutic agent for allergic contact dermatitis (ACD). In addition, these findings provide a basis for further investigation in the future such as molecular dynamics (MD) simulation, in vitro and in vivo experimental studies and clinical evaluation to further validate its efficacy and safety in inflammatory skin diseases.

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